

Full Length Article

Hybrid energy system as driver of sustainable rural development: An optimization and impact analysis of Tulefa Energy Village, Ethiopia

Sufyan Yakubu^{a,c,d,*}, Tsegaye Sissay Tedla^a, Sidiqie Gawusu^e, Samuel Dodobatia Wetajega^d, Mugisho Mugaruka Josue^f, Nasiru Alhassan^g, Getachew Adam Workneh^{a,b,*}

^a Sustainable Energy Center of Excellence, Addis Ababa Science and Technology University, P.O. Box 16417, Addis Ababa, Ethiopia

^b Department of Industrial Chemistry, Addis Ababa Science and Technology University, P.O. Box 16417 Addis Ababa, Ethiopia

^c Pan African University Institute of Water and Energy Sciences-University of Tlemcen, Tlemcen, Algeria

^d Department of Electrical and Electronic Engineering, Bolgatanga Technical University, Bolgatanga, Ghana

^e Whiting School of Engineering, Johns Hopkins University, Baltimore, MD, USA

^f Centre des Recherches en Environnement et Géo-Ressources, Université Catholique de Bukavu, DR, Congo

^g Department of Agricultural Engineering, Dr. Hilla Limann Technical University, Wa, Upper West Region, Ghana

ARTICLE INFO

Keywords:

HOMER

Hybrid energy

Sustainability

Rural electrification

Ethiopia

ABSTRACT

In this study, we investigated the design and optimization of a hybrid energy system for Tulefa Energy Village in Ethiopia using the HOMER software. The village is off-grid, with the majority relying on traditional biomass while facing huge energy challenges. The design and optimization consisted of solar photovoltaic, wind turbines, battery storage, and a diesel generator to deliver reliable and sustainable electricity. Four configurations were modelled and analyzed based on key parameters such as technical, economic, and environmental impacts. Among the analyzed configurations, the optimal system which consist of 35 kW solar photovoltaic, 5.1 kW wind turbine, 15 kW diesel generator and 91 kWh battery storage achieved renewable fraction of 98.8 % with a levelized cost of electricity of \$0.05018/kWh, and a total net present cost of \$68,363. Additionally, the system demonstrated minimal fuel consumption and greenhouse gas emissions, making it environmentally sustainable and economically viable. Sensitivity analysis was performed by changing variables such as discount rates, inflation rates, and fuel prices to assess the performance of the system. The results of this study indicate the potential of energy systems to drive rural development in Ethiopia by offering a scalable solution for off-grid communities. This adds to the international discourse on rural electrification, and therefore, to policies, engineering, and community development.

1. Introduction

Energy is a fundamental factor for economic and social development. It plays a vital role in improving quality of life, especially in the lives of developing societies. Areas where there is little or no access to energy, growth, and development is generally hampered. However, many developing countries are unable to provide their populations with reliable and sufficient electricity supply [1,2]. This situation is critical in Ethiopia, where most of the rural population depends on wood and charcoal obtained from biomass sources to meet their energy requirements. This overdependence results in environmental degradation through deforestation and serious health hazards owing to indoor air pollution [3].

These challenges have been addressed by integrating various renewable energy resources, such as solar photovoltaics, wind, and hydro, into hybrid systems, which have emerged as viable solutions. Such systems work effectively by exploiting the strengths of different energy sources to overcome their intermittency, which hinders the deployment of renewable energy [4]. Research findings indicate that hybrid systems can provide a more reliable and consistent power supply, making them particularly suitable for rural electrification in areas such as Ethiopia, where renewable resources are plentiful but not fully exploited.

Driven by concerns about climate change and global warming, the global installed capacity of renewable energy grew by 50 % in 2023 [5]. By the end of 2023, the global installed capacities of renewables,

* Corresponding authors.

E-mail addresses: yakubusuf@gmail.com (S. Yakubu), getachew.adam@aastu.edu.et (G.A. Workneh).

<https://doi.org/10.1016/j.nexus.2025.100434>

Received 23 October 2024; Received in revised form 26 February 2025; Accepted 22 April 2025

Available online 23 April 2025

2772-4271/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

including solar, wind, hydropower, geothermal, marine, and biogas, reached approximately 3372 GW [4]. This rapid expansion reflects a global shift towards renewable and sustainable energy technologies, with the renewable energy market projected to grow at an annual rate of 4.22 %. China and the United States lead this transformation, with installed capacities of 760 GW and 265 GW, respectively [5]. In Africa, where 221 GW of renewable energy capacity has been installed, hydropower, solar PV, and biomass dominate [6]. This growth highlights the global transition towards cleaner energy, which holds significant potential for improving rural electrification and sustainable development.

Previous studies have explored the design and optimization of energy systems in rural areas. For instance, Kifle et al. [7] explored the integration of solar, hydro, and biomass resources for rural electrification in the Jimma Zone, Ethiopia. This study found that optimized hybrid systems are cost-effective and reliable for providing energy to remote areas without grid access. A study by Tay et al. [8] on energy systems in rural Ghana demonstrated that integrating solar PV, wind, and diesel generators is both cost-effective and environmentally friendly. Using the HOMER software, the system was optimized, achieving a 55 % reduction in emissions compared to diesel-only setup. Research conducted by Girma [9] utilized HOMER software to optimize a hybrid renewable energy system for rural electrification in Ethiopia. The system incorporated photovoltaic panels, wind turbines, diesel generators, and battery storage components. The system achieved a high renewable energy fraction with diesel as a backup, reporting a levelized cost of energy (LCOE) of \$0.538/kWh and a Net Present Cost (NPC) of \$412,720. The study highlighted the system's cost-effectiveness and potential to improve energy access in rural areas. Similarly, Benti et al. [10] analyzed hybrid PV-diesel systems for rural schools, showing their economic viability and improved access to electricity through HOMER Pro optimization. Gezahegn et al. [11] explored hydro-wind-solar hybrid systems in Northern Ethiopia and found them to be technically feasible and economically advantageous for remote villages such as Seyemtribua. This approach highlights the potential of combining renewable sources to meet local energy demands. Furthermore, Beza et al. [12] compared various configurations of minigrid for the Kibran Gabriel island in Ethiopia using HOMER Pro software for techno-economic analysis. This study highlights the effectiveness of a PV/DG/battery system over diesel and grid extension options, with emphasis on lower costs and reduced emissions. The optimization of hybrid renewable energy systems is performed for rural Gaita Selassie, Ethiopia, using meta-heuristic algorithms, namely MOGWO and MOGOA, to obtain a minimum cost with maximum reliability for presenting an affordable and sustainable power supply for each underdeveloped community [13]. In Nigeria, Ijeoma et al. [14] assessed hybrid solar-battery-generator systems for supermarkets and showed that integrating these technologies significantly reduces LCOE while maintaining a reliable power supply, even with diesel price fluctuations. White et al. [15] assessed the techno-economic feasibility of a hybrid energy system in Grand Bassa, Liberia, using HOMER software. The optimal configuration, including biomass gasification, solar PV, and battery storage, proved cost-effective with a LCOE of \$0.29/kWh and an NPC of \$397,000. The study concluded that the system could significantly improve local socioeconomic conditions, particularly with external funding. B. Alqahtani et al. [16] investigated a hybrid renewable energy system combining pumped hydro storage, photovoltaics, and wind turbines, using a robust techno-economic analysis to assess the impact of head losses and evaporation rates on system performance; the findings indicate that incorporating these parameters improves the renewable energy fraction by approximately 8.6 % and reduces renewable energy loss by about 3 %, highlighting the importance of accurate modelling in optimizing system design. In a study by [17], the sensitivity of a hybrid renewable energy microgrid system integrating photovoltaics, wind turbines, diesel engines, and batteries, utilizing Typhoon HIL software for simulation and testing was investigated; the

findings indicate that a coordinated control strategy effectively enhances the system's stability and efficiency, providing significant insights into optimizing hybrid renewable energy resources in rural settings. The analyses of a hybrid photovoltaic-wind energy system integrated with pumped hydro storage using HOMER Pro software to assess its reliability and operational costs; the findings reveal that employing pumped hydro storage significantly reduces costs and operational issues compared to traditional battery systems, providing a viable solution for persistent energy supply in remote areas [18]. Furthermore, El-Khozondar et al. [19] assessed a hybrid off-grid energy system (HES) integrating photovoltaic (PV), wind, and diesel generators to power a COVID-19 quarantine center in Gaza, utilizing the HOMER-Pro software for modelling and optimization; the findings indicate that this HES can deliver cost-effective electricity at \$0.348/kWh, significantly reducing emissions by 54.89 % compared to diesel-only systems, while meeting the complete electrical load requirements efficiently. Moreover, J. O. Oladigbolu et al. [20] conducted a feasibility analysis of a hybrid renewable power system—combining micro-hydro, photovoltaic, wind, and diesel generators—using the HOMER simulation tool to determine the optimal design for off-grid rural electrification in a remote Nigerian village; the findings revealed that the hybrid system achieved a net present cost of \$1.01 million and a cost of energy of \$0.106/kWh, while significantly reducing environmental pollutant emissions compared to other configurations. Also, study by [21] assessed a dynamic analysis and optimization procedure for sizing a hybrid photovoltaic/wind energy system integrated with pumped hydroelectric storage (PHS) to ensure a sustainable electricity supply for an urban community in Brack City, Libya, employing a constrained optimization approach that accounts for renewable energy output variability; findings indicate that a power capacity ratio of 1:5 between solar PV and wind turbines is optimal, with PHS contributing 15 % to the annual energy load, demonstrating the system's cost-competitiveness and reliability for urban energy needs.

While previous studies have made significant progress in the design and optimization of hybrid energy systems for rural electrification, several critical gaps remain unaddressed. Most prior research focuses predominantly on techno-economic assessments, with limited consideration of the practical impacts of hybrid systems on rural development and socioeconomic outcomes. For instance, Girma et al. [9] and Beza et al. [12] have demonstrated the cost-effectiveness of hybrid systems incorporating solar PV, wind turbines, and diesel generators, but they have not fully explored the potential to enhance community development or improve quality of life. In addition, few studies have explored the integration of community driven initiatives or assess hybrid system in the context of experimental frameworks such as the Energy Village. Furthermore, the existing literature often overlooks the importance of sensitivity analyses to account for uncertainties such as fluctuating fuel prices, inflation rates, and discount rates. Also, the break-even distance or electric distance limit which compares the cost of traditional grid to proposed hybrid solution has also been overlooked in previous studies. This study fills that gap by not only optimizing the hybrid energy system for feasibility in socio-economic context, but also incorporating community participation in system design and evaluating its impact on rural development. The study conducts sensitivity analyses to account for uncertainties, ensuring the robustness of the system which has been overlooked by previous studies. Furthermore, the electric distance limit analysis is done to provide comparative evaluation with traditional grid offering insight into the cost effectiveness of hybrid energy system. By doing so, this study addresses these gaps by evaluating a hybrid energy system to assess its real-world impact of on rural development, particularly with a specific focus on community-driven initiatives like the Energy Village concept.

This is a concept under the Long-term Joint European Union–African Union Research and Innovation Partnership on Renewable Energy (LEAP-RE) Research and Innovation Action 963,530 consortium, which deals with the study of specific village renewable energy demands,

potential and efficient use, socioeconomic viability, and business potential. The LEAP-RE project introduced the concept of an Energy Village, which is a community-based approach for providing and demonstrating sustainable energy solutions in a rural setting. These Energy Villages represent experimentation spaces where a variety of renewable energy technologies are intended to be implemented and tested to obtain models that can then be replicated in other regions utilizing the localized energy resources of those locations. A good example is Tulefa Energy Village in Ethiopia, where this study builds on earlier work done on this location (LEAP-RE Horizon 2020 research and innovation programme WP14 Energy Village Concept in Africa).

This work augments the existing knowledge through an in-depth optimization and impact analysis for hybrid energy systems in Tulefa Energy Village, Ethiopia. In contrast to past studies that were centred around socio-economic impact and techno-economic evaluations, this study integrates self-help community initiatives as well as socio-economic impacts. The study optimizes the system for technical and economic feasibility and moreover uses sensitivity analyses to capture uncertainties such as inflation rates, discount rates, and fuel costs. A new addition in this study is the Energy Village concept which serves as a guideline for effective rural electrification and aims to provide practical solutions for policy makers, engineers, and community leaders. Additionally, this study is anticipated to enrich existing literature and conversations surrounding rural electrification and sustainable development, in Ethiopia and other countries.

2. Materials and methods

The modelled hybrid renewable energy system comprises a wind turbine, solar panels with a power converter, energy storage batteries, and a diesel generator. To evaluate the potential for renewable electricity generation in the village, the HOMER software employs climate databases from NASA and NREL. This assessment helps determine the viability of various renewable energy resources for the system. A load assessment was performed to determine the energy consumption and profile of the village. Input economic parameters are required and provided in the software. HOMER performed different configurations, and a comprehensive assessment of the economic, technical, and environmental sustainability analyses also performed on the different configurations to ascertain the most optimal system suitable for village electrification. Sensitivity analysis was performed to ascertain how changes in economic parameters, such as inflation rate, fuel price, and discount rate, affect output parameters, such as NPC, LCOE, and Capital Cost (CC). Additionally, the study conducted an Electric Distance Limit (EDL) analysis to compare the key indicators of the optimal solutions with the grid connection alternatives for the village.

2.1. Description of study area

Tulefa is located in the Amhara region of Ethiopia and northwest Shewa zone. It sits at an altitude of approximately 1805 m (5922 ft) above the sea level. Tulefa Energy Village is situated approximately 65 km from Addis Ababa in Ethiopia. The region features a striking landscape that is filled with rolling hills and valleys. It is abundant in underutilized renewable energy resources, such as solar, wind, and biomass. Located near the equator at 9°14'49" N, 39°14'59.9" E [22]. The village benefits from consistent solar energy for 12 h per day throughout the year. Moreover, wind is continuously present around the clock, and ample biomass resources offer significant potential for biogas production and power generation. The village is home to a small population of 250–300 people [23], with some densely populated households, whereas others are more scattered. Currently, the village is off-grid and relies mainly on firewood and biomass as primary energy sources. However, there is a significant potential for renewable energy in this area, particularly biomass, wind, and solar energy. Currently, none of the population is connected to the national grid [23]. Residents of

energy villages rely on livestock farming for commercial purposes and personal consumption.

2.2. Energy resource assessment

Energy resources can be assessed using various methods. Data can be obtained on the ground using on-site meteorological instruments or by assessing satellite meteorological data by the National Aeronautics and Space Administration (NASA) or the National Renewable Energy Laboratory (NREL) using HOMER software. The latter was used to assess solar and wind resources to eliminate the need for ground data (Fig. 1).

2.2.1. Global horizontal irradiance (GHI) and temperature

Data obtained indicates that the annual average Global Horizontal Irradiance (GHI) is 6.32 kWh/m²/day. As shown in Fig. 2(a), the monthly average GHI ranged from 5.935 kWh/m²/day in July to 6.730 kWh/m²/day in May. The skies tended to be clearer from September to June than they were from July to August. As depicted in Fig. 2(b), the annual average temperature is 25.05 °C with minimum and maximum temperatures in December at 22.20 °C and June at 28.050 °C respectively. The fraction of solar radiation from space that reaches the Earth's surface following air absorption and scattering is known as the clearness index (Kt). In solar energy studies, it is an essential metric since it helps in estimating solar radiation variability and determining whether photovoltaic (PV) systems are feasible in various geographic locations. Higher values (near 1.0) suggest bright skies with little air obstruction, while lower values (near 0.2–0.3) indicate cloudy or hazy conditions that limit the amount of direct solar radiation. The clearness index often ranges from 0 to 1. For Tulefa Energy Village, the annual average clearness index is estimated to be around 0.7, with monthly variations due to seasonal changes in cloud cover and atmospheric conditions. A high clearness index in the region suggests strong solar energy potential, making it favorable for photovoltaic (PV) power generation.

2.2.2. Wind speed

NASA's records indicate that the site's yearly mean wind velocity is 4.34 m/s when measured 20 m above the ground. As shown in Fig. 2(c), the monthly average wind speeds fluctuate, with the lowest being 3.51 m/s in April and the highest reaching 6.0 m/s in July.

2.3. Electrical load estimation/requirement

The electrical load estimation for Tulefa Energy Village was performed to ascertain the energy requirement of the community at large, both at the household and business levels, as well as in communal facilities. This is one of the most important steps in designing an appropriate and efficient energy system to meet the needs of the village.

A sample size of 40 people comprising 40 households (50 %) and business owners was interviewed to ascertain electricity requirements. In addition, community leaders' contributions were sought to identify the energy needs for public use in schools, community centre, and health centre.

The energy consumption for the area was estimated based on standard assumptions regarding load types, power demand, and usage duration, as assessed through a community survey. Equipment was sized using typical wattage ratings for common appliances, and a load assessment was conducted by analyzing the living and proposed consumption patterns of the occupants. The total energy consumption of the village was calculated using an Excel spreadsheet applying the formula: **Electrical energy (Wh) = Appliance power (W) × Usage time (h).**

The estimated daily energy demand at Tulefa Energy Village was around 104.35 kWh, taking into account the need for household appliances like lighting, television, and mobile phone chargers, among others, and community facilities like health centres, schools, police stations, churches and Mosques and others.

A summary of the calculated daily electricity requirements for the

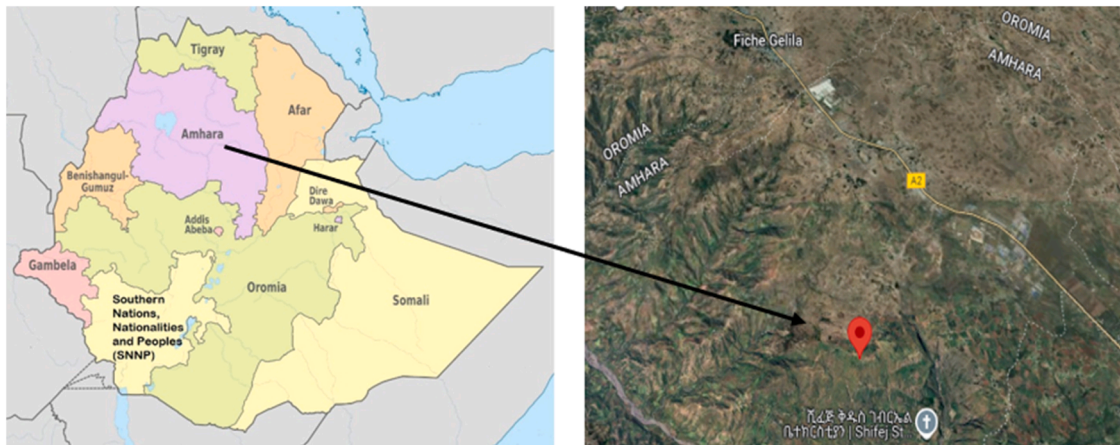
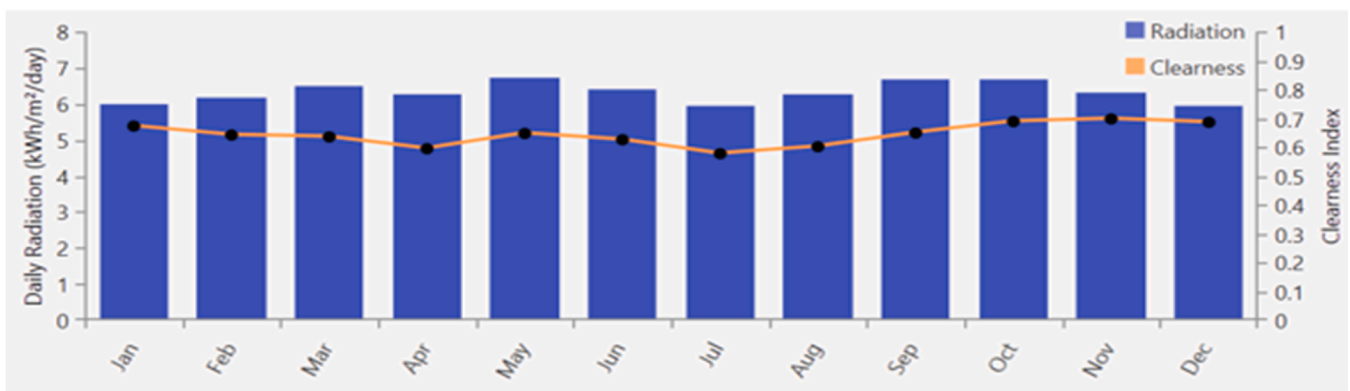
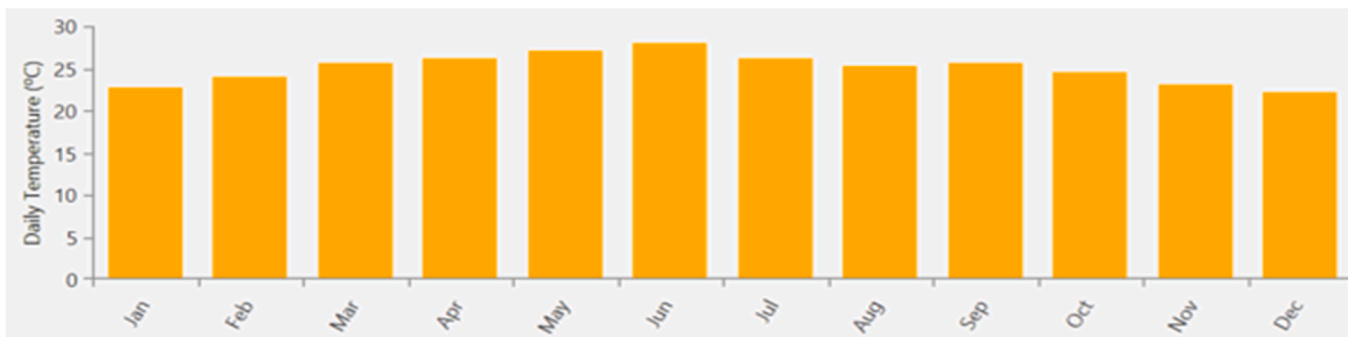


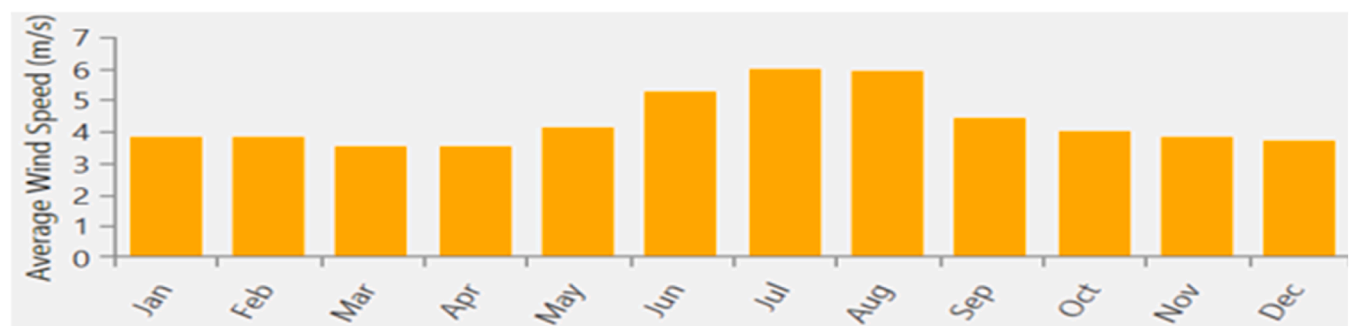
Fig. 1. Google earth view of study area [24].



(a)



(b)



(c)

Fig. 2. (a) Average Global Horizontal Irradiance of location (b) Average Temperature of location (c) Average Wind Speed of location [62].

villages is presented in Table 1.

2.4. HOMER software

HOMER software served as a tool for designing, simulating, optimizing, and analysing various power system configurations. These included off-grid and grid-connected systems, as well as stand-alone, remote, and distributed generation setups. It addresses both technical and financial aspects, allowing for the exploration of a wide range of technological solutions, while accounting for factors such as energy resource availability. HOMER identifies the most feasible system configuration that can reliably meet electricity demand, considering installation costs, replacement costs, operation and maintenance expenses, fuel costs, and interest rates (Fig. 3).

2.4.1. Assumptions, limitations, and constraints

The design and optimization of the proposed hybrid energy system are based on several key assumptions to streamline the modelling process. First, the study assumes that solar irradiance and wind speed data used in the simulations are consistent with historical averages, without accounting for potential deviations caused by extreme weather events or long-term climate changes. The energy demand of Tulefa Energy Village is considered to remain constant throughout the project's lifespan, disregarding possible fluctuations due to population growth or changes in consumption behaviour. Furthermore, the system components are modelled to operate under ideal conditions, without factoring in potential efficiency losses from degradation or unexpected maintenance needs.

Despite the robustness of the HOMER Pro model, this study has limitations that may affect its real-world applicability. The simplified load profile does not account for seasonal or economic changes in energy consumption, and the deterministic nature of HOMER Pro overlooks variability in factors like fuel prices and resource availability. Assumptions about battery and generator lifecycles may not fully reflect actual usage, while battery degradation over time could lead to higher operational and replacement costs than predicted. Additionally, challenges related to battery disposal and recycling raise environmental concerns. The study also excludes social and behavioural factors, such as shifts in energy usage patterns and community acceptance, which are crucial for the practical implementation of hybrid systems.

The constraints for the proposed hybrid energy system were established within HOMER Pro to achieve the optimal configuration tailored

to the needs of Tulefa Energy Village. The system was designed to fully meet the energy demand without any unmet load, ensuring reliable power supply. HOMER Pro optimization included a minimum renewable fraction constraint to maximize the use of renewable energy sources while minimizing reliance on diesel fuel. Resource availability, such as solar irradiance and wind speed, was modelled based on seasonal variability, ensuring realistic energy generation estimates. The battery storage system was constrained by safe state-of-charge limits to prevent over-discharge and maintain long-term efficiency. Similarly, the diesel generator was subject to runtime and efficiency constraints to minimize operational costs and greenhouse gas emissions. These constraints were systematically applied in HOMER Pro to identify the most cost-effective and sustainable hybrid system configuration that aligns with the community's energy needs and resource availability.

2.5. Component parameters

The sizes and capacities of solar panels and generators used in this study were determined based on the energy demand of Tulefa Energy Village and the availability of renewable resources in the region. The optimization process in HOMER Pro selected these specific capacities to meet the load requirements reliably while minimizing costs and maximizing renewable energy utilization. Additionally, market availability was a key consideration, ensuring that the proposed system components are practical and accessible for real-world implementation. As such, components information was gathered from the local market and HOMER software to accurately represent the average system cost. The selected capacities also account for future scalability, allowing the system to adapt to potential increases in energy demand over time. These factors were carefully considered to ensure the feasibility and sustainability of the proposed hybrid system.

2.5.1. Solar PV

Table 2 presents the financial and technical input data for the solar PV module utilized in the simulation. Apart from the rational that informed the general components of the system as highlighted in Section 2.5, the choice of PV was also hinged on regional solar radiations and environmental factors such as accumulation of dust which negatively affect the performance. Y. Nassar *et al.* highlighted the importance of choosing PV technology based on their electrical performance characteristics [25]. Aside market availability, a primary consideration for PV module selection were high yield, long-term durability and temperature coefficient. Mono-crystalline PV panels were preferred due to their high efficiency, better performance under moderate-to-high irradiance conditions making it suitable for Tulefa's climate.

2.5.2. Wind

Table 3 presents the chosen wind turbine, along with its associated expenses and technical specifications. The information used to determine the appropriate wind turbine was sourced from the component library within the HOMER software. These specifications and their corresponding values are typically consistent with those found in wind turbines available on the commercial market. The choice of turbine was based on site specific wind characteristics and terrain conditions at Tulefa Energy village. M. Elnaggar *et al.* explained the need of selecting wind turbine based on their high capacity and economic feasibility [27]. Tulefa experience moderate wind speed making it key to select turbine with low cut-in speed and high aerodynamic efficiency. Also, given the need of minimising maintenance in rural electrification projects, turbine with minimal mechanical wears were ideal.

2.5.3. Diesel generator

The technical and financial parameters of the Diesel Generator used in the simulation are listed in Table 4. The choice of a diesel generator was guided by its role in providing reliable energy particularly during low renewable generation. Energy demand at Tulefa must be met

Table 1
Overview of daily electricity requirement.

Appliance/ Category	Power Rating (W)	Quantity	Daily Usage (hours)	Daily Energy Consumption (Wh)
Household Lighting	10	150	5	7500
Televisions	75	30	5	11,250
Radios	15	50	6	4500
Mobile Phone Charging	5	200	2	2000
Water Pumps	500	2	4	4000
Community Centre Lighting	50	10	6	3000
Street Lighting	40	20	12	9600
Small Business Appliances	200	10	8	16,000
Health Centre Equipment	300	5	8	12,000
Police station	100	10	24	24,000
School Computers	150	10	4	6000
Other Miscellaneous Loads	100	15	3	4500
				Total = 104,350

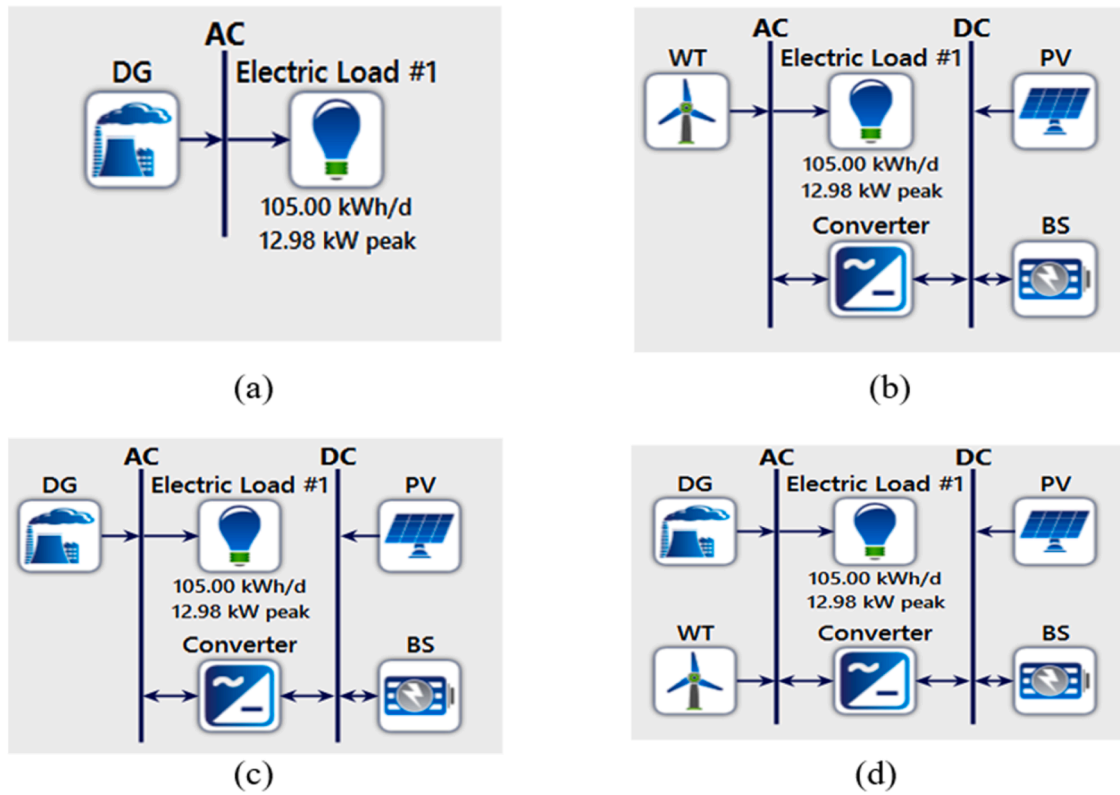


Fig. 3. (a) Option 1 (b) Option 2 (c) Option 3 (d) Option 4.

Table 2
PV module parameters.

Manufacturer	Canadian Solar	Reference
Item no.	CS6W-535MB-AG	
Type	Monocrystalline	
Rated Max. Power (W)	535	
Voltage at Pmax (V)	41.1	[26]
Current at Pmax (I)	13.02	
Efficiency (%)	20.8	
Lifetime (Years)	25	
Capital Cost per kW (\$)	450	Market data in Ethiopia
Yearly O&M Cost per kW (\$/year)	5	Authors Assumption

Table 3
Wind turbine parameters.

Make	AWS	Reference
Item no.	AWS HC 5.1 kW Wind Turbine	
Rated Power (W)	51,000	
Cut in m/s	2.7	[28,29]
Rated wind speed m/s	11	
Survival wind speed	55 m/s	
Hub height	20	
Lifetime (Years)	20	
Capital Cost (\$)	5000	
Replacement cost (\$)	3000	Market survey data in Ethiopia
Yearly O&M Cost per kW (\$/year)	50	and Authors assumption

consistently and due to the intermittency nature of renewable energy sources, a generator with high fuel efficient, low maintenance requirement and fast response time was chosen to complement the hybrid energy system.

Table 4
Parameters of diesel generator.

Manufacturer/Model	CAT C2.2	Reference
Fuel Type	Diesel	
Capital Cost (\$/kW)	500	
Replacement Cost (\$)	500	Market data in Ethiopia
Cost of diesel (\$/liter)	0.75	
O&M Cost (\$/hr)	0.03	
Lifespan (hours)	20,000	[30,31]

2.5.4. Converter

The converter parameters used in the simulations are presented in Table 5. The choice of inverter was guided efficiency, adaptability to variable generation from different sources and compatibility with hybrid energy systems. The role of advanced Maximum Power Point Tracking (MPPT) algorithms in energy conversion efficiency improvement in hybrid energy systems has been emphasized. With intermittent nature of solar and wind energy resources, an inverter with multi-MPPT functionality was chosen to optimize power generation.

2.5.5. Battery storage

The technical and financial input parameters for battery storage used in this study are listed in Table 6. The choice of battery guided on the

Table 5
Converter parameters.

Manufacturer/Model	Fronius Symo 15.0-3-M	Reference
Type	Hybrid Converter	
Efficiency (%)	90	[32]
Battery Voltage (VDC)	48	
Lifetime (year)	13	
Capital Cost (\$/kW)	100	
Replacement Cost (\$/kW)	90	Market data in Ethiopia

Table 6

Battery parameters.

Manufacturer/Model	Generic	Reference
Type	Lithium-ion	
Rated Voltage (V)	6	
Battery capacity (Ah)	167	
Nominal Capacity (kWh)	1	
Efficiency (%)	90	[33]
State of Charge (Initial) (%)	100	
State of Charge (Minimum) (%)	20	
Throughput (kWh)	3000	
Lifetime (years)	15	
Unit Cost (\$)	180	
Replacement Cost (\$/kW)	150	Local Market data

emphasis of high cycle life and efficiency hence lithium-ion batteries with high depth of discharge (DoD), better efficiency and longer lifespan were preferred.

2.6. Economic input parameters

HOMER software utilizes economic parameters such as the inflation rate and discount rate for the simulation. Trading Economics global macro models and analysts' expectations quoted the expected end-of-year inflation rate of Ethiopia to be 18 % and projected inflation to trend around 14.5 % and 13 % in 2025 and 2026, respectively [34]. However, the National Bank of Ethiopia has quoted a discount rate of 15 % [35].

Sensitivity analysis, often referred to as "what if" is typically conducted to assess how changes in an input variable affect an output variable. Inputs such as the inflation rate, discount rate, and fuel cost were selected for the analysis, while output variables such as NPC, CC, LCOE, and operation and maintenance (O&M) costs were monitored. The sensitivity analysis parameters are presented in Table 7.

2.7. Mathematical model equations

2.7.1. Net present cost (NPC)

$$NPC = \sum_{t=1}^n \frac{Rt}{(1+i)^t} \quad (1)$$

Where: Rt is the net cash inflow/outflow during a single period t, i is the discount rate, and t is the number of periods [36].

2.7.2. Levelized cost of energy (LCOE)

$$LCOE = \frac{CRF * CAPEX + OPEX}{Et} \quad (2)$$

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (3)$$

Where: CAPEX is the capital expenses, OPEX is the operational expenses, Et is the electricity produced in year t, CRF is the capital recovery factor, N is the project lifetime, and i is the interest rate [37,38].

2.7.3. Annualized cost

$$Cann = CRF * NPC \quad (4)$$

Table 7

Input economic and sensitivity values.

Input Parameter	Rate	Sensitivity value
Expected Inflation rate	18 %	13 %, 14.5 %
Discount rate	15 %	10 %, 7 %
Diesel price	\$0.75	\$1 \$1.2

Where:

Cann is the annualized cost, CRF () is the Capital Recovery factor function, as indicated in Eq. (3), and NPC is Net Present Cost, as in Eq. (1) [39].

2.7.4. Salvage value (SV)

$$SV = P(1 - r)^n \quad (5)$$

Where: P is the original price, r is the depreciation rate, n is the age in years, and SV is the salvage value [40].

2.7.5. Real discount rate

$$i = \frac{(a - f)}{(1 + f)} \quad (6)$$

Where: a is the nominal discount, f the expected inflation rate, and i the real discount [41].

2.7.6. Wind energy

$$E_W = \begin{cases} P_{rat} \left(\frac{V_{Z,t} - V_{cut-in}}{V_{rat} - V_{cut-in}} \right) V_{cut-in} < V_{Z,t} < V_{cut-off} \\ 0 & V_{Z,t} \leq V_{cut-in} \text{ OR } V_{Z,t} \geq V_{cut-off} \end{cases} \quad (7)$$

Where: P_{rat} is the rated power of the wind turbine at rated wind speed V_{rat} , V_{cut-in} and $V_{cut-off}$ are the cut-in and cut-off wind speeds, and $V_{Z,t}$ is the wind speed at the wind turbine hub height (h_z) and it is calculated from [42]:

$$V_{Z,t} = V_{0,t} \left(\frac{h_z}{h_0} \right)^\alpha \quad (8)$$

where: $V_{0,t}$ is the wind speed at a certain elevation (h_0) and α is the wind shear coefficient [43]

2.7.7. The real power (P_{PV}) of the PV panel

$$P_{PV} = P_{STC} [1 + \beta_p (T_{cell} - T_{STC})] \frac{H_t}{H_{STC}} \quad (9)$$

Where: T_{STC} and H_{STC} are standard test conditions of temperature [$^{\circ}\text{C}$] and solar radiation [Watt/m^2] respectively, β_p is the power temperature coefficient [44].

$$T_{cell} = T_{\infty} + 7.8 \times 10^{-2} H_t \quad (10)$$

Where: H_t is the solar radiation intensity [Watt/m^2], T_{∞} and T_{cell} are the ambient air and the surface cell temperatures [$^{\circ}\text{C}$], respectively [45].

2.7.8. Environmental damage by CO_2

The cost of generated energy unit (LCOE) and the payback time money (PBTM). The cost of environmental damage produced by carbon dioxide (C_{CO_2}) may be used to compute LCOE using the following equation:

$$LCOE = \frac{\left(\frac{r(1+r)^n}{(1+r)^n - 1} \right) \times C + C_{O\&M} - C_{\text{CO}_2}}{E_t} \quad (11)$$

$$PBTM = \frac{C}{\frac{(1+r)^n - 1}{r(1+r)^n} (C_{\text{CO}_2} - C_{O\&M})} \quad (12)$$

where: C the capital cost of the system in \$, $C_{O\&M}$ denotes the cost of

operation and maintenance (\$/year), E_t is annual energy produced by the system (kWh/year), n is the device lifetime (30 years), r the annual inflation rate [46].

Environmental impact of energy systems is assessed by quantifying CO_2 associated with generated energy. Emissions are directly proportional to fuel consumption in convectional diesel generator systems. The CO_2 emission from fuel combustion can be calculated using the following equation:

$$E_{\text{CO}_2} = F \times EF_{\text{CO}_2} \quad (13)$$

Where: E_{CO_2} is the total CO_2 emissions (kg CO_2), F is the total fuel consumed (liters) and EF_{CO_2} is the emission factor (kg CO_2 per liter of diesel). This is suitable diesel fuel only system [47].

However, in hybrid energy systems, the emission is influenced by both fuel consumption and energy generation requiring a more refined approach. The cost of environmental damage (C_{CO_2}) caused by CO_2 gas can be calculated by the following equation:

$$C_{\text{CO}_2} = EF_{\text{CO}_2} \times E_t \times \phi_{\text{CO}_2} \quad (14)$$

where: EF_{CO_2} represents the CO_2 emission factor of the electric power generation system (kg CO_2/kWh), ϕ_{CO_2} represents the carbon social cost (\$/ton CO_2), which may be considered as \$ 70/ton CO_2 and E_t represents the total electrical energy generated by the system over specific period (kWh) [48,49].

3. Results and discussion

HOMER simulated and optimized four configured scenarios, as presented in Table 8.

The schematics of the optimized options are presented in Fig. 4.

HOMER simulated and optimized the system designs to consistently meet the energy demand of the village and hence, come in different configured sizes. The PV and WT were designed to meet taken into consideration the intermittency requirement. The DG is sized to meet the continuous and peak demand of the village and BS to store and dispatch excess energy from generation sources. This ensures that the system maintains reliability even under extreme conditions, such as prolonged periods of low solar irradiance or wind speeds. HOMER considers these factors when sizing system components, which explains why elements such as PV, WT, converters, and BS differ in size, while the DG remains consistent across all configurations.

This integrated design not only enhances reliability by ensuring a continuous power supply but also improves system efficiency by reducing reliance on the diesel generator and optimizing renewable energy utilization. As a result, the system achieves high renewable fractions and lower operational costs, making it both economically and environmentally sustainable. Table 9 presents a comparison of the indicators of various scenarios, and a graphical representation is indicated in Fig. 5.

The DG Only (O1) system has the highest Net Present Cost of \$727,170, which reflects its reliance on diesel fuel and high operational costs. On the other hand, the PV + WT + BS (O2) architecture significantly reduces the NPC to \$90,199 owing to the absence of fuel costs and

the higher use of renewable energy systems. The PV + DG + BS (O3) and PV + WT + DG + BS (O4) systems also show reductions in NPC at \$73,888 and \$68,363, respectively, highlighting the cost-effectiveness of integrating more renewable components.

O1 has the lowest capital cost of \$7500, which is expected, given its simpler architecture with no renewable energy components. O2 showed a higher capital investment of \$53,369, mainly because of the integration of wind turbines, solar panels, and battery storage. Systems such as O3 have an initial capital cost of \$48,410, while the cost of O4 is \$45,991. Thus, although hybrid systems require more upfront investment, the benefit lies in savings elsewhere.

The replacement costs also follow similar trends: O1 has the highest replacement cost of \$105,118 because of the frequent replacement of the diesel generator. On the other hand, the renewable-based system O2 has a very low replacement cost of \$36,776, while for O3 and O4 systems, the replacement costs have further been reduced to estimated costs of \$25,504 and \$26,907, respectively.

Accordingly, the O&M cost of \$140,124 was the highest for O1, driven by the maintenance required for diesel generators. Indeed, renewable-based systems such as O2 reduce O&M costs drastically to \$20,150. Furthermore, O3 and O4 reduced the cost to \$17,385 and \$16,211, respectively.

Fuel cost is a key differentiating factor among the systems. O1 relies entirely on diesel, leading to a high fuel cost of \$475,142. Systems such as O2 completely eliminate fuel costs, whereas O3 and O4 have minimal fuel costs of \$5795 and \$5306, respectively, owing to the limited diesel backup.

The O1 system had the highest LCOE value of \$0.5338/kWh, primarily because of its reliance on high fuel and operational costs. However, the O2 system has a lower LCOE of \$0.06624/kWh, making it cost-effective. Additionally, hybrid systems, such as O3 and O4, offer even more affordable energy solutions, with LCOE values of \$0.05424 and \$0.05018, respectively. This demonstrates that the integrated hybrid systems can provide more economically feasible energy solutions.

The ranking from highest to lowest includes O2 at 106,654 kWh/year, O3 at 84,787 kWh/year, and O4 at 74,246 kWh/year. O1 generates the least electricity among all the systems at 44,402 kWh/year, further showing weaknesses depending on pure diesel power.

In contrast, O2 has the highest amount of excess electrical generation, with an excess of 63,999 kWh/year. The fact that the potential for surpluses in generation characterizes all renewable-based systems means that the two systems O3 and O4 may produce excess power of 40,064 and 30,729 kWh/year, respectively, which may be exported or reserved. The DG-Only-based system generates the least excess of electricity at 6077 kWh/year.

O1 has no contribution from renewable sources, and the renewable fraction is 0 %. In O2, there is 100 % renewable energy, while in O3, it is 98.7 % and 98.8 % for O4, respectively, indicating very good integration due to renewable energies.

The fuel consumption has a maximum by far from O1, which is 17,822 liters/year, and the maximum emission of greenhouse gases is 47,351 kg/year. O2, on its part, showed no fuel consumption or emissions. However, for O3 and O4, fuels decreased to 217 and 199 liters per year, while the GHG emissions were 578 and 528 kg/year, respectively.

The obtained results illustrate that hybrid systems integrated with renewable energy components minimize operational costs, fuel consumption, and greenhouse gas emissions and enhance efficiency and sustainability. The DG Only system presents the minimum capital cost but is highly expensive in terms of fuel and maintenance costs; therefore, it is not feasible in the long run. Among these options, the PV + WT + BS (O2) and PV + WT + DG + BS (O4) systems provide the best balance of cost-effectiveness, environmental friendliness, and reliability for remote or off-grid areas.

Among these, the O4 system had the most optimal configuration. While maintaining a high renewable energy fraction of 98.8 %, it includes a diesel generator for backup, ensuring a reliable power supply,

Table 8
Configurations.

Scenario	Optimized Configuration
Option 1 (O1)	15 kW Diesel Generator (DG) only (Base Case)
Option 2 (O2)	48.2 kW Photovoltaic (PV) + 10.2 kW Wind Turbine (WT) + 113 kWh Battery Storage (BS)
Option 3 (O3)	45 kW PV + 15 kW Diesel Generator (DG) + 106 kWh Battery Storage (BS)
Option 4 (O4)	35 kW Photovoltaic (PV) + 5.1 kW Wind Turbine (WT) + 15 kW Diesel Generator (DG) + 91 kWh Battery Storage (BS)

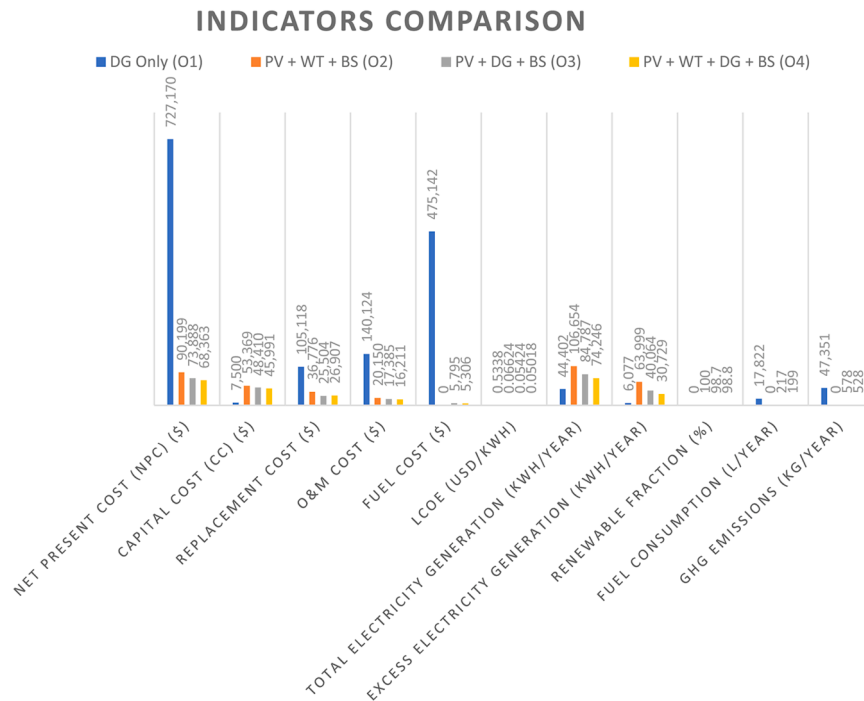


Fig. 4. Graphical comparison of indicators.

Table 9
Summary of key indicators of the scenarios.

Parameter	DG Only (O1)	PV + WT + BS (O2)	PV + DG + BS (O3)	PV + WT + DG + BS (O4)
Net Present Cost (NPC) (\$)	727,170	90,199	73,888	68,363
Capital Cost (CC) (\$)	7500	53,369	48,410	45,991
Replacement Cost (\$)	105,118	36,776	25,504	26,907
O&M Cost (\$)	140,124	20,150	17,385	16,211
Fuel Cost (\$)	475,142	0	5795	5306
LCOE (USD/kWh)	0.5338	0.06624	0.05424	0.05018
Total Electricity Generation (kWh/year)	44,402	106,654	84,787	74,246
Excess Electricity Generation (kWh/year)	6077	63,999	40,064	30,729
Renewable Fraction (%)	0	100	98.7	98.8
Fuel Consumption (L/year)	17,822	0	217	199
GHG Emissions (kg/year)	47,351	0	578	528

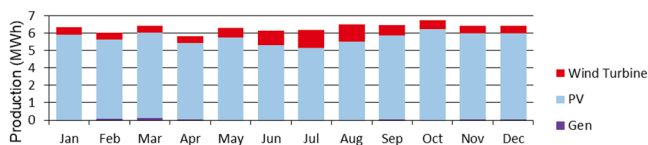


Fig. 5. Optimum system power production.

even during periods of low solar and wind availability. This feature, combined with its lower NPC of \$68,363 and LCOE of \$0.05018/kWh, makes it economically advantageous over time.

Furthermore, the system emits only 528 kg of GHG per year, a significant reduction compared to the DG Only (O1) alternative. With its minimal fuel consumption and high efficiency, the PV + WT + DG + BS (O4) hybrid system stands out as the most reliable, cost-effective, and

environmentally sustainable option for long-term energy needs for Tulefa Energy village.

3.1. The optimal system

Configuration PV + WT + DG + BS, on that account, is the most optimal since it strikes a good balance between two renewable energy systems, namely solar photovoltaic and wind turbines, and backup with a diesel generator to secure continuity of power supply in times of low availability of solar radiation or wind. The Levelized Cost of Energy (LCOE) for the optimized PV + WT + DG + BS system is \$0.05018/kWh, which is the lowest among all configurations analyzed. This low LCOE is primarily attributed to the high renewable energy fraction (98.8 %), significantly reducing the reliance on diesel fuel, which is expensive and subject to price volatility. The integration of solar PV and wind turbines ensures a consistent energy supply, while battery storage minimizes the need for diesel generator usage, further lowering operational costs. The system architecture is presented in Table 10, Figs. 5 and 6 shows system power production and the performance of the various components respectively.

This result is significant because it demonstrates the economic competitiveness of hybrid energy systems compared to traditional diesel-only solutions. For instance, the LCOE of the DG-only system (\$0.5338/kWh) is over ten times higher, highlighting the cost-saving potential of renewable energy integration. Furthermore, this LCOE aligns with global trends in declining renewable energy costs, particularly for solar PV and battery storage, making hybrid systems increasingly viable for rural electrification projects in developing countries.

Table 10
Optimum system architecture.

Component	Specification
Photovoltaic Array	35 kW
Wind Turbine	5.1 kW
Converter	14 kW
Battery Storage	91 kWh
Diesel Generator	15 KW

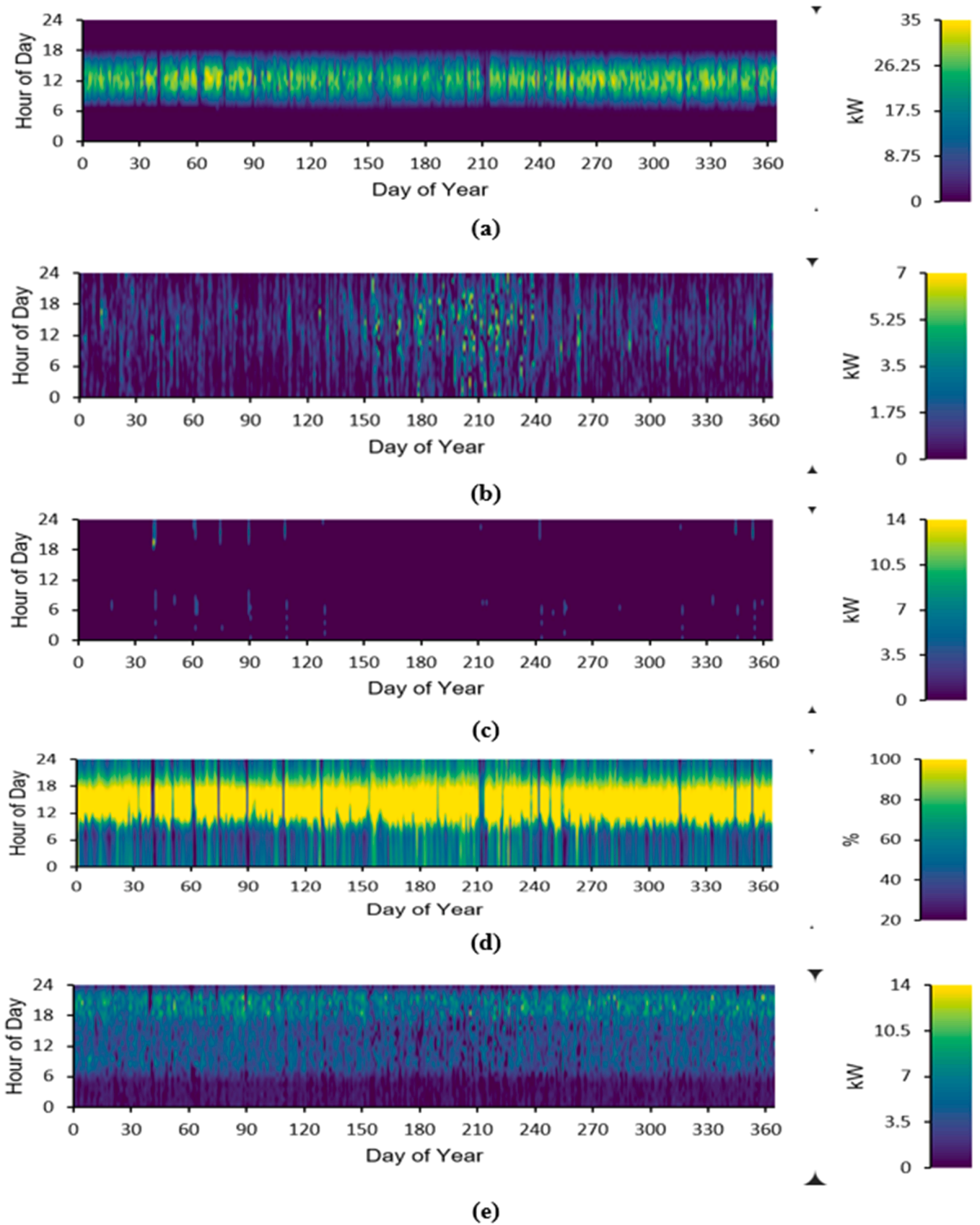


Fig. 6. Power output of optimum components (a) PV (b) Wind (c) Diesel Generator (d) Battery Storage (e) Converter.

The Net Present Cost (NPC) of the optimized PV + WT + DG + BS system is \$68,363, significantly lower than other configurations. This low NPC is achieved through reduced operational and maintenance (O&M) costs due to the minimized use of diesel fuel and efficient system sizing. By leveraging HOMER Pro's optimization capabilities, the system components, including battery storage and wind turbine, are sized to balance initial capital costs with long-term operational savings.

The NPC's significance lies in its reflection of the system's overall cost-effectiveness. For comparison, the DG-only configuration exhibits NPC of \$727,170, demonstrating the financial burden of reliance on fossil fuels. The lower NPC of the hybrid system underscores its potential to provide affordable and sustainable energy for rural communities. Additionally, it highlights the importance of optimizing system components to achieve economic feasibility in real-world applications.

The optimized PV + WT + DG + BS system achieves a renewable fraction of 98.8 %, with minimal reliance on diesel generators. This is reached through smartly combining solar, wind, batteries and a small generator backup. Photovoltaics, turbines, and stored power addressed intermittency together. The intermittent energy sources generated the vast majority, with the diesel only occasionally supplying scarce extra when natural sources provided little. This highly renewable mix demonstrates effective integration of technologies to minimize fossil fuel dependency.

While rural Ethiopian communities continue facing energy access obstacles reliant on biomass, this result proves particularly impactful. By lowering dependence on fossil fuels, the hybrid model cuts annual greenhouse gas by 528 kg while easing health and environmental challenges tied to conventional biomass usage. The high renewable fraction also demonstrates the technical feasibility of achieving near-complete renewable reliance, contributing to Ethiopia's broader sustainability goals.

From the simulation, the optimum system may produce excess power 30,729 kWh/year representing 41 % of the system energy output which may be exported and sold to nearby communities. The excess energy can be utilized to meet the future household expansion needs of the village. It could also act as a catalyst for economic activities by boosting the establishment of more local businesses as the lack of electricity is hindering local business development. Other critical communities load infrastructure could be powered by the excess energy and all these can be realized by understanding the load profile of the village and implementing efficient energy management practices.

3.1.1. Comparative environmental impact

Renewable energy integration into the hybrid energy systems greatly decreases environmental impact, especially in terms of GHG emission and fossil fuel dependency. The contribution of carbon dioxide (CO₂), nitrous oxide (N₂O) and particulate matter (PM) emissions to air pollution and climate change by convectional diesel generator system only is high.

From HOMER optimised results, the standalone diesel generator only systems produce 47,351 kg of GHG per year whereas the optimised hybrid system reduces diesel runtime by 99 % leading the estimated 99 % reduction in emissions. This aligns with findings from previous finding such as [50,51] which report similar emission reduction in hybrid energy systems.

3.2. Sensitivity analysis

The sensitivity analysis shows the response of both NPC and LCOE to changes in some key economic parameters like discount rate, fuel price, and inflation rate.

LCOE is greatly affected by the discount rate, as it determines the present value of future costs and benefits of the hybrid system. Higher discount rates put more weight on up-front capital costs and less on long-term operational savings. For instance, as the discount rate increases from 7 % to 15 %, the LCOE in this study rises sharply from \$0.00615/

kWh to \$0.05018/kWh as depicted in Fig. 7, reflecting the system's reduced cost-effectiveness under higher discount rates. This sensitivity underscores the importance of financing terms and policies in determining the economic viability of renewable energy projects. In contrast, lower discount rates enhance the attractiveness of hybrid systems by spreading costs more evenly over the system's lifetime.

Fuel price also exerts a noticeable impact on LCOE, albeit less pronounced than the discount rate. In hybrid systems with diesel backup, such as the one analyzed, rising fuel prices increase the operational costs of the system. However, the relatively small contribution of diesel to the overall energy mix in the optimized configuration (PV + WT + DG + BS) mitigates the extent of this impact. For example, even as fuel prices increase from \$0.75/L to \$1.2/L as shown in Fig. 8, the LCOE experiences only a modest change, stabilizing at higher values. This highlights the resilience of systems with a high renewable energy fraction to fuel price volatility.

Inflation has a less direct but still significant effect on LCOE, as it influences the costs of system components, maintenance, and fuel over time. For example, increasing the rate from 13 % to 18 % brings the NPC from \$62,157 to \$68,363, while the LCOE declines from \$0.0808/kWh to \$0.05018/kWh as depicted in Fig. 9. It just goes to prove that inflation has a very strong influence on overall costs. The higher the rate of inflation, the higher the NPC and the lower the LCOE.

This analysis indicates that economic parameters, particularly the discount rate, have a more significant impact on LCOE than technical factors such as resource availability. This is because economic variables dictate the financial attractiveness of the project over its lifetime, while technical factors primarily influence system performance. Additionally, hybrid systems with high renewable energy penetration, such as PV + WT + DG + BS, are less sensitive to fluctuations in fuel prices due to their minimal dependence on diesel fuel and robust renewable energy contribution.

3.3. Comparison with previously published literature

The optimal system from this study is compared to previous research with similar studies. Economic output parameters tend to vary across studies due to factors such as system components, size, cost, and energy resource availability. However, The LCOE is frequently employed as a standard measure for comparing various studies as it directly impacts end-use affordability. Table 11 highlights the comparison between this study and previous works based on the LCOE indicator.

The findings show that the current study has a lower LCOE than previous studies, which can be attributed to the ongoing decline in renewable energy costs, making it increasingly viable for rural electrification, especially in areas far from the grid.

To validate the hybrid energy system further, comparative analysis with other alternative energy sources is explored. One example of alternative energy source explored for rural electrification is biomass-

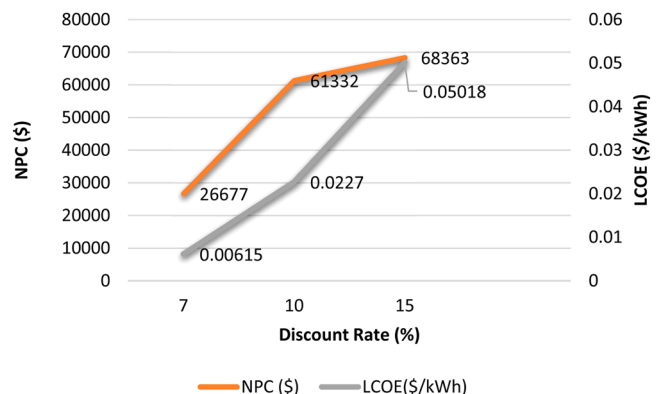


Fig. 7. Impact of discount rate on NPC and LCOE.

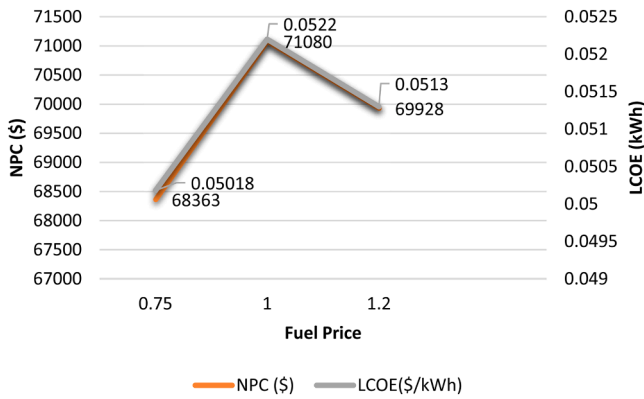


Fig. 8. Impact of fuel price on NPC and LCOE.

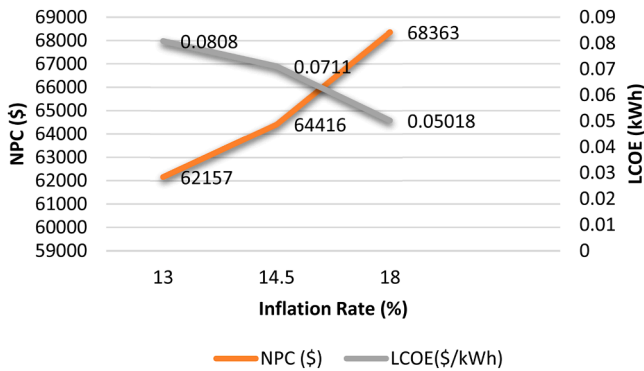


Fig. 9. Impact of Inflation rate on NPC and LCOE.

Table 11
Comparison with previous literature.

Reference	Location	Configuration	Status	LCOE (\$/kWh)
[52]	Indonesia	PV + WT + BS	Off-grid	0.24
[53]	Ghana	PV + DG + BS	Off-grid	0.399
[54]	Egypt	PV + BS + Biogas Generator (BG)	Off-grid	0.14
[41]	India	PV + WT + BS	Off-grid	0.278
[55]	Kenya	PV + WT + BG + BS	Off-grid	0.25
[56]	Ethiopia	PV + WT + DG + BS	Off-grid	0.207
[57]	Ethiopia	PV + WT + BS + Fuel Cell (FC)	Off-grid	0.313
Present Study	Ethiopia	PV + WT + DG + BS	Off-grid	0.05018

based power generation especially areas with abundant organic waste. While Tulefa possess ample biomass resources, technical complexities and seasonal variation usually hinders widespread adoption. Additional, study on biomass power generation by [58] reports LCOE ranging from US\$0.29/kWh and US\$0.34/kWh which is higher remains higher than the optimised.

Furthermore, small scale hydropower is one energy source that can provide cost effective energy. A study by [59] highlighted that hydropower is site-dependent and requires specific hydrological conditions. Given Tulefa lacks a major rivers or water flow suitable for power generation, hydropower is not a viable option. Extensive comparative analysis of grid extension option is extensively explained in Section 3.5.

Comparative analysis with other energy sources confirms the optimized hybrid system is the most suitable solution offering low LCOE, minimal environmental impact and high reliability.

3.4. Comparison with real-world applications

The simulation results demonstrate the economic and technical viability of hybrid energy systems for rural electrification. However, their alignment with real-world applications is influenced by various factors.

3.4.1. Alignment with real-world applications

The LCOE of \$0.05018/kWh and NPC of \$68,363 obtained in this study align with reported values in similar projects, particularly those implemented in sub-Saharan Africa. For instance, previous studies have shown that hybrid systems integrating solar PV, wind, and battery storage offer cost-effective solutions, often achieving LCOE values below \$0.5/kWh in rural settings. These results underscore the feasibility of hybrid systems for providing affordable energy to underserved communities.

The decline in renewable energy technology costs, particularly solar PV and battery storage, further supports the alignment of simulation results with real-world trends. In addition, the inclusion of a diesel generator for backup ensures reliability, addressing intermittency challenges often observed in rural deployments.

3.4.2. Deviations from real-world applications

While the simulations provide optimized results under controlled parameters, real-world applications are subject to uncertainties that may cause deviations. Factors such as fluctuating fuel prices, variations in renewable resource availability, unexpected maintenance requirements, and delays in project implementation can impact the actual costs. For example, the simulation assumes stable solar and wind resource availability, which may vary seasonally in practice. Additionally, the initial capital cost for hybrid systems may be higher in remote areas due to logistical challenges and limited infrastructure.

Another key deviation lies in social and behavioral factors. The model assumes full utilization of the system's capacity, but actual usage patterns may differ due to user behavior or cultural practices. Furthermore, policy and regulatory uncertainties, such as tariffs and subsidies, can affect the financial feasibility of hybrid systems.

3.4.3. Implications for real-world applications

To bridge the gap between simulation and real-world scenarios, it is essential to incorporate risk management strategies, conduct sensitivity analyses, and consider local socio-economic contexts during project planning. The findings of this study emphasize the importance of ongoing monitoring and adaptive management to ensure that hybrid systems meet their intended performance and cost-effectiveness in practice.

3.5. Electric distance limit for grid extension

Electric Distance Limit (EDL) or Break-even Distance refers to the maximum distance at which it becomes more economically feasible to implement a renewable energy-based off-grid power system rather than extending an electrical grid to a remote area. In other words, it refers to the point where the NPC of extending the grid matches the NPC of a standalone energy system [60] estimated the cost of extending 11 KV grid line in sub-Saharan Africa to be \$25,000/km and could be more for worse terrains and mountainous areas. For the purpose of this study, we have adjusted the estimated by 40 % considering the difficult terrain of the area and to cater for inflation to achieve \$35,000/km. The break-even distance for this study as depicted in Fig. 10 is found to be 1.95 km which means that connecting Tulefa energy village to the conventional grid is only beneficial if the distance between the source

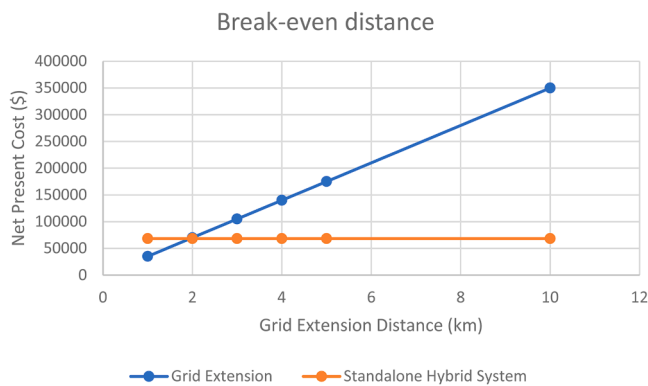


Fig. 10. Break-even distance.

substation and Tulefa is 1.95 km or less.

In Ethiopia, 55 % of the population has access to electricity, while the figure drops to 43 % for those living in rural areas [61]. The nearest grid point is approximately 25 km away from the area, and given that most regions without electricity in Ethiopia are located hundreds of kilometres from the grid, extending the grid would be significantly more expensive than implementing the suggested standalone system. The continuous decrease in prices of renewable energy prices and the continuous increase in grid extension make standalone system cost competitive and further supports microgrid for electrification of hard-to-reach areas far away from the grid.

4. Conclusion

The enhanced design of the Tulefa Energy Village hybrid system showcases the practicality and potential economic viability of incorporating renewable energy sources in rural settings. Several different configurations have been evaluated, ranging from a Diesel Generator-only configuration to various hybrid combinations, including PV, WT, BS and DG, and all prove that hybrid systems are much more economical and environmentally viable than traditional DG only systems.

Among these configurations evaluated, the PV + WT + DG + BS system is selected for the most optimal solution in Tulefa. This setup achieves an optimal equilibrium between incorporating sustainable energy sources and maintaining system dependability by combining solar and wind capabilities, along with a diesel generator for backup support. This setup reaches a renewable energy fraction of 98.8 %, while maintaining a LCOE of \$0.05018/kWh, the lowest among the alternatives, hence making the system most economically viable. It also has NPC of \$68,363, emitting only 528 kg of greenhouse gas emissions per year with the least fuel consumption. The sensitivity analysis further confirms the robustness of this system, showing resilience to uncertainties in key economic parameters. Additionally, the break-even distance supports the use of hybrid stand-alone systems as an effective tool for electrifying off-grid areas far from the grid network. With government incentives and tax exemptions, renewable energy-based microgrids will become even more cost-effective, resulting in the lowest possible LCOE and NPC Policymakers, governments, and stakeholders should consider this approach to accelerate energy access in remote areas.

This study underlines that renewable energy-based hybrid systems provide cost savings, enhance energy security, and reduce environmental impacts. Surplus energy generation capacity from the system opens perspectives for future expansion and economic development in Tulefa. Added to this are the decreasing costs of renewable technologies and increasing grid extension costs, further reinforcing the case for stand-alone systems such as the one proposed in Tulefa, becomes even stronger, especially in remote or off-grid areas.

This study presents a novel approach to rural electrification by

integrating solar photovoltaics, wind turbines, batteries, and diesel generator into a single system that is optimized towards Tulefa Energy Village. Unlike other studies, this research analyses community needs and looks at the socio-economic impacts of the system, rather than focusing only on techno-economic feasibility analysis. The results show that the hybrid system not only serves as a cost-efficient environmentally friendly energy source, but also meets the needs of other off-the-grid communities. These findings add to the discussion on rural electrification by showing how hybrid energy systems can help achieve sustainable development in isolated areas. Adding on to these findings, future works should focus on applying this model to other regions and looking in depth into socio-economic advantages of such systems.

CRedit authorship contribution statement

Sufyan Yakubu: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tsegaye Sissay Tedla:** Visualization, Resources, Data curation. **Sidique Gawusu:** Writing – review & editing, Writing – original draft, Validation, Conceptualization. **Samuel Dodobatia Wetajega:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Mugisho Mugaruka Josue:** Validation, Software, Resources, Conceptualization. **Nasiru Alhassan:** Validation, Conceptualization, Resources, Writing – review & editing. **Getachew Adam Workneh:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported by the Intra-Africa Academic Mobility Scheme in the framework of the project [624175-PANAF-1-2020-1-ET-PANAF-MOBAF, Electrochemical Science and Innovative Materials for Sustainable Africa Development (ESIMSAD)] and 963530 — LEAP-RE — H2020-LC-SC3-2018-2019-2020/H2020-LC-SC3-2020-JointActions-

Acknowledgements

The authors gratefully acknowledge the funding from ESIMSAD. The authors will also like to acknowledge the LEAP-RE Horizon 2020 research and innovation programme WP14 Energy Village Concept in Africa, and the Sustainable Energy Centre of Excellence of Addis Ababa Science and Technology University for their useful support to this work.

Data availability

Data used to support the findings of this study are available from the corresponding author upon request.

References

- [1] F. Carlsson, E. Demeke, P. Martinsson, T. Tesemma, Cost of power outages for manufacturing firms in Ethiopia: a stated preference study, *Energy Econ* 88 (2020) 104753, <https://doi.org/10.1016/j.eneco.2020.104753>.
- [2] S. Yakubu, R. Samikannu, J. Lydia, L.S. Vimalraj, R. Karpagam, A. Yahya, Exploring the progress of decentralized photovoltaic systems in rural Ghana: a comprehensive review, 2023 IEEE Renew. Energy Sustain. E-Mobility Conf. RESEM (2023) 1–6, <https://doi.org/10.1109/RESEM57584.2023.10236139>, 2023.
- [3] A.H. Nebey, Design of optimal hybrid power system to provide reliable supply to rural areas of Ethiopia using MATLAB and Homer, *Renew. Wind. Water, Sol.* 8 (1) (2021), <https://doi.org/10.1186/s40807-021-00067-w>.
- [4] IRENA, Renewable capacity statistics, 2024. [Online]. Available, www.irena.org, 2024.

- [5] International Energy Agency, Renewables 2020, Paris (2020), <https://doi.org/10.1002/peng.20026>.
- [6] Renewable energy potential in the State of Palestine, Proposals for sustainability - ScienceDirect, Accessed, <https://www.sciencedirect.com/science/article/abs/pii/S1755008424000401>, Dec. 28, 2024.
- [7] F. Kifle, V. Ramayya, G. Shunki, A. Beyene, Potentiality and feasibility study of a hybrid renewable energy system for remote area electrification in Jimma Zone, Int. J. Nov. Res. Electr. Mech. Eng. 10 (2022) 85–99, <https://doi.org/10.5281/zenodo.8131195>, July.
- [8] G. Tay, et al., Optimal sizing and techno-economic analysis of a hybrid solar PV/wind/diesel generator system, IOP Conf. Ser. Earth Environ. Sci. 1042 (1) (2022), <https://doi.org/10.1088/1755-1315/1042/1/012014>.
- [9] Z. Girma, Hybrid renewable energy design for rural electrification in Ethiopia, J. Energy Technol. Policy 3 (13) (2013) 38–53 [Online]. Available, <http://www.iiste.org/Journals/index.php/JETP/article/viewFile/9765/9972>.
- [10] N.E. Benti, et al., Techno-economic analysis of solar energy system for electrification of rural school in Southern Ethiopia, Cogent Eng 9 (1) (2022) 1–21, <https://doi.org/10.1080/23311916.2021.2021838>.
- [11] G.T. Gezahegn, S.D. Hatiye, M.T. Ayana, A.T. Ayalew, and T.T. Minda, *The hybrid power generation and balancing energy supply-demand for rural village in Ethiopia*, vol. 47, no. 1. 2023. doi: 10.1111/opec.12264.
- [12] T.M. Beza, C.H. Wu, C.C. Kuo, Optimal sizing and techno-economic analysis of minigrid hybrid renewable energy system for tourist destination islands of lake tana, ethiopia, Appl. Sci. 11 (15) (2021), <https://doi.org/10.3390/app11157085>.
- [13] E.F. Agajie, et al., Optimization of off-grid hybrid renewable energy systems for cost-effective and reliable power supply in Gaita Selassie Ethiopia, Sci. Rep. 14 (1) (2024) 1–26, <https://doi.org/10.1038/s41598-024-61783-z>.
- [14] M.W. Jjeoma, C.G. Lewis, H. Chen, B.N. Chukwu, M. Carbajales-Dale, Technical, economic, and environmental feasibility assessment of solar-battery-generator hybrid energy systems: a case study in Nigeria, Front. Energy Res. 12 (2024) 1–18, <https://doi.org/10.3389/fenrg.2024.1397037>, May.
- [15] J.G. White, et al., Techno-economic feasibility assessment and performance analysis of standalone solar photo voltaic-biomass hybrid system with optimized storage: a case study—Grand Bassa, Front. Energy Res. 12 (2024) 1–15, <https://doi.org/10.3389/fenrg.2024.1326558>, March.
- [16] B. Alqahtani, J. Yang, M.C. Paul, Design and performance assessment of a pumped hydro power energy storage connected to a hybrid system of photovoltaics and wind turbines, Energy Convers. Manag. 293 (2023) 117444, <https://doi.org/10.1016/j.enconman.2023.117444>, October.
- [17] M.G.M. Almihat, "Solar energy," vol. 0, pp. 80–104, 2023.
- [18] H. Koli, M.P.S. Chawla, Design analysis of PV-wind energy system with pumped hydro storage using HOMER Pro, Int. J. Recent Technol. Eng. 9 (3) (2020) 502–505, <https://doi.org/10.35940/ijrte.c4597.099320>.
- [19] H.J. El-Khozondar, F. El-batta, R.J. El-Khozondar, Y. Nassar, M. Alramlawi, S. Alsadi, Standalone hybrid PV/wind/diesel-electric generator system for a COVID-19 quarantine center, Environ. Prog. Sustain. Energy 42 (3) (2023) 1–18, <https://doi.org/10.1002/ep.14049>.
- [20] J.O. Oladigbolu, M.A.M. Ramli, Y.A. Al-Turki, Feasibility study and comparative analysis of hybrid renewable power system for off-grid rural electrification in a typical remote village located in Nigeria, IEEE Access 8 (2020) 171643–171663, <https://doi.org/10.1109/ACCESS.2020.3024676>, April 2023.
- [21] Y.F. Nassar, et al., Dynamic analysis and sizing optimization of a pumped hydroelectric storage-integrated hybrid PV/wind system: a case study, Energy Convers. Manag. 229 (December 2020) 113744, <https://doi.org/10.1016/j.enconman.2020.113744>, 2021.
- [22] Tulofa map - locality - Oromiya, Ethiopia, Accessed, <https://mapcarta.com/13000174>, Sep. 02, 2024.
- [23] I. Action, E. Union, A. Haapanen, and P.A. Kiprop, "Report on background analysis for pilot villages as deliverable of task 14. 2," 2021.
- [24] 9°14'49.0"N 39°14'59.9"E - Google Maps, Accessed, https://www.google.com/maps/place/9°14'49.0%22N+39°14'59.9%22E/@9.21287,39.3136467,12421a,35y,270h,37.61t/data=!3m1!1e3!4m3!8m2!3d9.2469444!4d39.2499722!5m1!1e4?entry=ttu&g_ep=EgoyMDI0MDg5OC4wIXkxMDSoASAFQAw%3D%3D, Sep. 03, 2024.
- [25] Y.F. Nassar, H.J. El-Khozondar, S.Y. Alsadi, N.M. Abuhumoud, G.M. Miskeen, Atlas of PV solar systems across Libyan territory, in: Proc. - 2022 Int. Conf. Eng. MIS, ICEMIS 2022, 2022, pp. 1–6, <https://doi.org/10.1109/ICEMIS56295.2022.9914355>.
- [26] Canadian Solar, Preliminary technical information Sheet 520 W ~ 550 W CS6W-520[525][530][535][540][545][550]MB-AG (IEC1000 V) [Online]. Available, www.csisolar.com/au, 2021.
- [27] M. Elnaggar, H.J. El-Khozondar, W.A. Salah, Y.F. Nassar, M.J.K. Bashir, Assessing the Techno-enviro-economic viability of wind farms to address electricity shortages and foster sustainability in Palestine, Results Eng 24 (2024) 103111, <https://doi.org/10.1016/j.rineng.2024.103111>, October.
- [28] Z. Arifin, et al., Grid-connected hybrid PV-wind system simulation in urban Java, J. Eur. des Syst. Autom. 55 (4) (2022) 477–483, <https://doi.org/10.18280/jesa.550406>.
- [29] N. Santino, et al., Feasibility study of a standalone hybrid energy system to supply electricity to a rural community in South Sudan, Sci. African 16 (2022) e01157, <https://doi.org/10.1016/j.sciaf.2022.e01157>.
- [30] "Generator." Accessed: Feb. 23, 2025. [Online]. Available: <https://homerenergy.com/products/pro/docs/3.15/generator.html>.
- [31] C2.2 Industrial Diesel Engines | Cat | Caterpillar, Accessed, https://www.cat.com/en_US/products/new/power-systems/industrial/industrial-diesel-engines/18391931.html, Feb. 23, 2025.
- [32] Symo Advanced 15.0-3-M, Accessed, <https://www.fronius.com/en-au/australia/solar-energy/installers-partners/technical-data/all-products/inverters/fronius-symo-advanced/symo-advanced-15-0-3-m>, Feb. 23, 2025.
- [33] Library. Storage, Accessed, https://homerenergy.com/products/pro/docs/3.15/sstorage_library.html, Feb. 23, 2025.
- [34] Ethiopia Inflation Rate, Accessed, <https://tradingeconomics.com/ethiopia/inflation-cpi>, Sep. 09, 2024.
- [35] NBE Sets Policy Rate at 15 % in Shift to Interest-Rate-Based Monetary Policy - Capital Newspaper, Accessed, <https://www.capitalethiopia.com/2024/07/09/nbe-sets-policy-rate-at-15-in-shift-to-interest-rate-based-monetary-policy/>, Sep. 09, 2024.
- [36] Net Present Value (NPV): What It Means and Steps to Calculate It, Accessed, <https://www.investopedia.com/terms/n/npv.asp>, Sep. 06, 2024.
- [37] J. Langer, J. Quist, K. Blok, Recent progress in the economics of ocean thermal energy conversion: critical review and research agenda, Renew. Sustain. Energy Rev. 130 (December 2019) 109960, <https://doi.org/10.1016/j.rser.2020.109960>, 2020.
- [38] D.A. Quansah, S.K. Woangbah, E.K. Anto, E.K. Akowuah, M.S. Adaramola, Techno-economics of solar pv-diesel hybrid power systems for off-grid outdoor base transceiver stations in Ghana, Int. J. Energy a Clean Environ. 18 (1) (2017) 61–78, <https://doi.org/10.1615/InterJEnerCleanEnv.2017019537>.
- [39] Total Annualized Cost, Accessed, https://homerenergy.com/products/pro/docs/3.15/total_annualized_cost.html, Feb. 13, 2025.
- [40] M.B. Kadarmydeen, M. Kwarteng, Feasibility design, comparative evaluation, and energy consumption prediction of standalone hybrid energy system for rural electrification in Ghana using machine learning models, Results Eng (2025) 104269, <https://doi.org/10.1016/j.rineng.2025.104269>.
- [41] P.K. Sahu, S. Jena, U. Sahoo, Techno-economic analysis of hybrid renewable energy system with energy storage for rural electrification, Hybrid Renew. Energy Syst. (2021) 63–96, <https://doi.org/10.1002/9781119555667.ch3>.
- [42] Y. Fathi Nassar, S. Yassin Alsadi, Wind energy potential in Gaza Strip-Palestine state, Sol. Energy Sustain. Dev. J. 7 (2) (2021), <https://doi.org/10.51646/jesd.v7i2.73>.
- [43] A. Abdalla, W. El-Osta, Y.F. Nassar, W. Husien, E.I. Dekam, G.M. Miskeen, Estimation of dynamic wind shear coefficient to characterize best fit of wind speed profiles under different conditions of atmospheric stability and terrains for the assessment of height-dependent wind energy in Libya, Appl. Sol. Energy (English Transl. Geliotekhnika) 59 (3) (Jun. 2023) 343–359, <https://doi.org/10.3103/S0003701X23600212/METRICS>.
- [44] A.A. Hafez, Y.F. Nassar, M.I. Hamdan, S.Y. Alsadi, Technical and economic feasibility of utility-scale solar energy conversion systems in Saudi Arabia, Iran. J. Sci. Technol. - Trans. Electr. Eng. 44 (1) (Mar. 2020) 213–225, <https://doi.org/10.1007/s40998-019-00233-3/METRICS>.
- [45] Y.F. Nassar, A.A. Salem, The reliability of the photovoltaic utilization in southern cities of Libya, Desalination 209 (1–3) (Apr. 2007) 86–90, <https://doi.org/10.1016/j.desal.2007.04.013>.
- [46] A. Elmariami, W. El-Osta, Y. Nassar, Y. Khalifa, M. Elfleet, Life cycle assessment of 20 MW wind farm in Libya, Appl. Sol. Energy (English Transl. Geliotekhnika) 59 (1) (Feb. 2023) 64–78, <https://doi.org/10.3103/S0003701X22601557/METRICS>.
- [47] UNDP. Co 2 Emission Reduction Calculation , Standardized Baseline Emission Factor Setting, and Mrv in the Building Sector Under the Paris Agreement, 2019.
- [48] A.M. Makhzom, et al., Carbon dioxide life cycle assessment of the energy industry sector in Libya: a case study, Int. J. Electr. Eng. Sustain. 1 (3) (2023) 145–163 [Online]. Available, <https://ijees.org/index.php/ijees/article/view/58>.
- [49] M.A. Eteriki, W.A. El-Osta, Y.F. Nassar, H.J.E. Khozondar, Effect of implementation of energy efficiency in residential sector in Libya, 8th Int. Eng. Conf. Renew. Energy Sustain. ieCRES (2023) 2023, <https://doi.org/10.1109/IECRES57315.2023.10209521>.
- [50] H. Al-Rawashdeh, et al., Performance analysis of a hybrid renewable-energy system for green buildings to improve efficiency and reduce GHG emissions with multiple scenarios, Sustain 15 (9) (2023), <https://doi.org/10.3390/su15097529>.
- [51] E. Niringiyimana, S. Wanquan, G. Dushimimana, J.B. Niyigena, Hybrid renewable energy system design and optimization for developing countries using HOMER Pro: case of Rwanda, in: 2023 7th Int. Conf. Green Energy Appl. ICGEA 2023, 2023, pp. 72–76, <https://doi.org/10.1109/ICGEA57077.2023.10125739>.
- [52] A. Xu, L.J. Awalin, A. Al-Khaykan, H.F. Fard, I. Alhamrouni, M. Salem, Techno-economic and environmental study of optimum hybrid renewable systems, including PV/wind/gen/battery, with various components to find the best renewable combination for Ponorogo regency, Sustain 15 (3) (2023), <https://doi.org/10.3390/su15031802>.
- [53] A.K. Awopone, Feasibility analysis of off-grid hybrid energy system for rural electrification in Northern Ghana, Cogent Eng 8 (1) (2021), <https://doi.org/10.1080/23311916.2021.1981523>.
- [54] M.B. Eteiba, S. Barakat, M.M. Samy, W.I. Wahba, Optimization of an off-grid PV/biomass hybrid system with different battery technologies, Sustain. Cities Soc. 40 (2018) 713–727, <https://doi.org/10.1016/j.scs.2018.01.012>.
- [55] S.G. Sigarchian, R. Paleta, A. Malmquist, A. Pina, Feasibility study of using a biogas engine as backup in a decentralized hybrid (PV/wind/battery) power generation system - case study Kenya, Energy 90 (2015) 1830–1841, <https://doi.org/10.1016/j.energy.2015.07.008>.
- [56] K. Gebrehiwot, M.A.H. Mondal, C. Ringler, A.G. Gebremeskel, Optimization and cost-benefit assessment of hybrid power systems for off-grid rural electrification in Ethiopia, Energy 177 (2019) 234–246, <https://doi.org/10.1016/j.energy.2019.04.095>.

- [57] M.H. Kebede, G.B. Beyene, Feasibility study of PV-wind-fuel cell hybrid power system for electrification of a rural village in Ethiopia, *J. Electr. Comput. Eng.* 2018 (2018), <https://doi.org/10.1155/2018/4015354>.
- [58] F. Odoi-Yorke, L.K. Osei, E. Gyamfi, M.S. Adaramola, Assessment of crop residues for off-grid rural electrification options in Ghana, *Sci. African* 18 (2022) e01435, <https://doi.org/10.1016/j.sciaf.2022.e01435>.
- [59] T. Ohshima, "T_{μγ}:: status and prospects," no. May 2001, pp. 327–337, 2002, doi: [10.1142/9789812776488_0034](https://doi.org/10.1142/9789812776488_0034).
- [60] O.M. Longe, K. Ouahada, H.C. Ferreira, S. Chinnappen. Renewable Energy Sources microgrid design for rural area in South Africa, *ISGT 2014*, Washington, DC, USA, 2014, pp. 1–5. [10.1109/ISGT.2014.6816378](https://doi.org/10.1109/ISGT.2014.6816378).
- [61] |. Ethiopia, S.D.G. Tracking, 7." Accessed, <https://trackingsdg7.esmap.org/country/ethiopia>, Sep. 16, 2024.
- [62] HOMER - Hybrid Renewable and Distributed Generation System design Software, Accessed, <https://homerenergy.com/products/pro/index.html>, Dec. 29, 2024.